

NexaCare Solutions: Testing and Release Planning Kit

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Project: Appointment Scheduling Platform – Version 3.0 Release

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Project scenario

NexaCare Solutions is a healthcare technology company supporting over 300 clinics across North America. The organization is preparing to launch Version 3.0 of its web-based appointment scheduling platform.

This version introduces new features that aim to improve usability, patient engagement, and clinic operations. As the Systems Analyst, your responsibility is to coordinate and document a complete testing and release plan to ensure a safe, compliant, and successful rollout.

Key features in version 3.0

- SMS reminders
- Real-time doctor availability
- Calendar integration with Google and Outlook
- Redesigned user interface

Key considerations

- HIPAA compliance (over 1.2M patient records)
- First-time integration with third-party SMS provider
- Diverse client configurations (from basic to API-integrated systems)
- Pilot program for 20 clinics prior to full release
- Scheduled deployment window: Saturday, 12:00 AM – 6:00 AM EST

1. Test Strategy Outline

Overall Approach: A risk-based, multi-level testing strategy prioritizing data integrity and third-party integration reliability.

Types of Testing:

Unit Testing: Verify individual code modules for SMS logic and calendar calculations.

Integration Testing: Crucial for the new third-party SMS provider and Google/Outlook APIs.

Regression Testing: Ensure existing 300+ clinics retain full functionality of basic scheduling.

Performance/Load Testing: Simulate Monday morning peak (2500 concurrent users) to ensure <3s booking times.

UAT: Conducted by the 20 pilot clinics to validate real-world workflows and the redesigned UI.

Test Priorities: 1. Data Integrity (Migration), 2. SMS Delivery, 3. Real-time Availability Sync.

Objectives: System stability, HIPAA compliance, and maintaining <2m SMS delivery latency.

2. Traceability Matrix

Requirement ID	Requirement Description	Test Case ID(s)	Coverage Status	Priority / Risk
REQ-SMS	SMS Reminders (24h/1h)	TC-SMS-01, TC-SMS-02	Covered	High (New Provider)
REQ-AVAIL	Real-time Doc Availability	TC-AVAIL-01	Covered	High (Concurrency)
REQ-CAL	Google/Outlook Sync	TC-CAL-01	Covered	Medium (API Dep)
REQ-UI	WCAG Redesigned UI	TC-UI-01	Covered	Medium (Usability)
REQ-DATA	HIPAA/Data Integrity	TC-MIG-01	Covered	Critical (Data Loss Risk)

3. Annotated Test Plans

Feature: SMS Reminders

Test Objective: Validate that reminders are sent exactly 24h and 1h before appointment.

Preconditions: Active third-party SMS API key; appointments scheduled in test DB.

Step 1: Schedule appointment for T+25 hours. **Expected:** 24h SMS triggers at T+1h.

Annotation/Risk: Third-party Latency. If API response is >2m, it violates NFR. Monitor for rate-limiting during bulk sends.

Feature: Real-time Doctor Availability

Test Objective: Ensure update reflects across all sessions within 5s.

Step 1: Provider cancels a 10:00 AM slot. **Expected:** Slot disappears from patient view instantly.

Annotation/Risk: Race Conditions. High risk of double-booking if two patients select the same slot within the 5s sync window.

4. Deployment Checklist

Pre-deployment Tasks

Full Backup: Database Admin to take a point-in-time snapshot of 1.2M records (Friday 11:30 PM).

Pilot Ready: Confirm 20 pilot clinics have login credentials for Version 3.0.

Deployment Window (Sat 12:00 AM – 6:00 AM)

12:00 AM: Put system in maintenance mode; stop writes.

12:30 AM: Execute data migration scripts.

03:00 AM: Deploy application code; run automated smoke tests.

05:00 AM: Go/No-Go Decision. Milestone: All automated tests pass and record counts match.

5. Rollback Strategy

Triggering Conditions: Any data loss detection, critical bug in booking logic, or system latency >10s during smoke tests.

Procedure: 1. Point DNS back to the legacy environment. 2. Restore Database from 11:30 PM snapshot if migration corrupted data.

Communication: Notify support teams and clinics via internal dashboard within 15m of rollback initiation.

Est. Rollback Time: 45 – 90 minutes.

6. Data Migration Plan

Migration Volume: 1.2M Patient Records (Health & Contact data).

Field Mapping Sample:

Legacy_Phone \$\\rightarrow\$ SMS_Target: Format to E.164 (+1XXXXXXXXXX).

Provider_Schedule \$\\rightarrow\$ Live_Availability: Map to new real-time schema.

Validation Steps: * **Pre-migration:** Sanity check for null phone numbers.

Post-migration: 100% record count verification and field-level checksums.

7. Post-Deployment Monitoring Summary

Metric	Threshold / SLA	Monitoring Tool
Uptime	99.9% Availability	CloudWatch / Synthetic Canaries +1
SMS Latency	< 2 Minutes	Custom Logs / API Monitoring +1
Error Rate	< 0.5% HTTP 5xx	Application Performance Monitor (APM) +1

8. Non-Functional Requirements & Performance Plan

A. NFR Definition

Performance: Appointment booking completion < 3 seconds (Measured via API response time).

Scalability: Support 2500 concurrent users during Monday peak (Measured via Load testing).

Latency: Availability updates reflect within 5 seconds (Measured via Client-side telemetry).

B. Golden Signals Monitoring

Latency: API P95 response time; Target < 500ms.

Traffic: Active websocket connections; Target < 2500.

Errors: Rate of failed SMS sends; Threshold < 1%.

Saturation: Database CPU utilization; Threshold < 80%.

C. Peak Traffic Preparation (Monday 8–10 AM)

Infrastructure: Trigger auto-scaling groups to add 2 extra application nodes at 7:30 AM EST.

Alerting: Adjust Database CPU alert threshold from 85% to 95% to avoid false alarms during expected heavy load.